

PAPER_505: MSVC

Release-MaxCompress Build Profile

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026
Source: PAPER_505_MSVC_Release_MaxCompress_Build_Profile.md

PAPER_505: MSVC Release-MaxCompress Build Profile

Session: 137 | **Source:** grok_share_84a767d3.txt (lines 1500–1900, 2700–2800) **Date:** November 2025 — verified commit 146a3c4 **Machine:** Trident01 — AMD Ryzen 5 5600G, 128 GB RAM, Windows 11

§1.1 Abstract

The Release-MaxCompress build profile is the canonical optimized production configuration for MAIN_1_CoAnQi.cpp under MSVC (Visual Studio 2022). It combines whole-program optimization, link-time code generation, size-favoring code generation, and post-build UPX compression to produce the smallest possible fully-functional x64 executable. This paper documents each flag, its effect, and the achieved result: **1.75 MB** for a 108,000-line C++20 monolith.

§1.2 Complete Flag Table

Category	Flag	Effect	Size/Speed Gain
Standards	/std:c++20	Full C++20 conformance	—
Standards	/permissive-	Strict standards mode	—
Standards	/Zc:__cplusplus	Correct __cplusplus macro value	—
Architecture	/arch:AVX2	256-bit AVX2 SIMD (Ryzen 5 5600G)	+10–20% speed

Category	Flag	Effect	Size/Speed Gain
Optimization	<code>/GL</code>	Whole Program Optimization	WPO enabled
Optimization	<code>/Os</code>	Favor small code size	−15–25% size
Optimization	<code>/Gw</code>	Global data optimization	−10–20% size
Optimization	<code>/GF</code>	String pooling	minor
Optimization	<code>/Gy</code>	Function-level linking (COMDAT)	−5–15% size
Optimization	<code>/Oi</code>	Enable intrinsic functions	+5–15% speed
Linker	<code>/LTCG</code>	Link-Time Code Generation	WPO finalization
Linker	<code>/OPT:REF</code>	Remove unreferenced code/data	−20–30% size
Linker	<code>/OPT:ICF</code>	Identical COMDAT folding	−5–10% size
Post-build	UPX <code>--best</code> <code>--ultra-brute</code>	Runtime decompressor	−70% size

§1.3 CMake Implementation

```

if(MSVC)
    # Base C++20 setup
    add_compile_options(/permissive-)
    add_compile_options(/Zc:__cplusplus)

    # Release-MaxCompress flags (only in Release config)
    add_compile_options("$&lt;$&lt;CONFIG:Release>&gt;:/arch:AVX2>&quot;)
    add_compile_options("$&lt;$&lt;CONFIG:Release>&gt;:/GL>&quot;)
    add_compile_options("$&lt;$&lt;CONFIG:Release>&gt;:/Os>&quot;)
    add_compile_options("$&lt;$&lt;CONFIG:Release>&gt;:/Gw>&quot;)
    add_compile_options("$&lt;$&lt;CONFIG:Release>&gt;:/GF>&quot;)
    add_compile_options("$&lt;$&lt;CONFIG:Release>&gt;:/Gy>&quot;)
    add_compile_options("$&lt;$&lt;CONFIG:Release>&gt;:/Oi>&quot;)
    add_link_options("$&lt;$&lt;CONFIG:Release>&gt;:/LTCG>&quot;)
    add_link_options("$&lt;$&lt;CONFIG:Release>&gt;:/OPT:REF>&quot;)
    add_link_options("$&lt;$&lt;CONFIG:Release>&gt;:/OPT:ICF>&quot;)
endif()

# Optional post-build UPX compression
find_program(UPX_EXECUTABLE upx)
if(UPX_EXECUTABLE)
    add_custom_command(TARGET MAIN_1_CoAnQi POST_BUILD

```

```

COMMAND ${UPX_EXECUTABLE} --best --ultra-brute
    &quot;${&lt;TARGET_FILE:MAIN_1_CoAnQi&gt;}&quot;;
COMMENT &quot;UPX compressing executable&quot;;
)
endif()

```

§1.4 Compression Chain Analysis

Starting from uncompressed debug build:

```

Debug x64 (no opt)           → ~50 MB
+ /GL /LTCG /OPT:REF /OPT:ICF → ~17.5 MB (-65%)
+ /Os                        → ~13.3 MB (-24%)
+ /Gw /GF /Gy               → ~11.3 MB (-15%)
+ /arch:AVX2 /Oi            → ~11.0 MB (-2%, mainly speed)
= Pre-UPX Release binary    → ~11.0 MB
+ UPX --best --ultra-brute   → ~1.75 MB (-84%)
= FINAL MAIN_1_CoAnQi.exe    → 1.75 MB ■

```

§1.5 Build Commands

```

# Configure (Visual Studio 17 2022, x64)
cmake -S . -B build_msvc -G &quot;Visual Studio 17 2022&quot; -A x64

# Build Release target
cmake --build build_msvc --config Release --target MAIN_1_CoAnQi

# Clean rebuild
Remove-Item -Recurse -Force build_msvc -ErrorAction SilentlyContinue
cmake -S . -B build_msvc -G &quot;Visual Studio 17 2022&quot; -A x64
cmake --build build_msvc --config Release

```

§1.6 Verification

After build, verify:

```

(Get-Item &quot;build_msvc\Release\MAIN_1_CoAnQi.exe&quot;).Length / 1MB # Should be ~1.75
cmake --build build_msvc --config Release 2&gt;&amp;1 | Select-String &quot;error C|Finished&quot;;

```

§1.7 Equations

Compression ratio $C_ratio = size_final / size_debug$

LTCG gain	$G_{\text{LTCG}} = (\text{size_no_LTCG} - \text{size_with_LTCG}) / \text{size_no_LTCG}$
AVX throughput factor	$F_{\text{AVX2}} = \text{throughput_AVX2} / \text{throughput_SSE2} \approx 2 \times (\text{256-bit width})$
UPX expansion ratio	$R_{\text{UPX}} = \text{time_decompression} / \text{time_cold_start} \quad (\text{\< 50 ms typical})$

§1.8 Citation

Source: grok_share_84a767d3.txt, lines 1500–1900, 2700–2800 Confirms: 1.75 MB executable at commit 146a3c4 (Nov 22, 2025) Machine: AMD Ryzen 5 5600G, 128 GB RAM, Windows 11, MSVC v14.44.35219
Paper number: PAPER_505